

S V V Sai Sri Vallabh Pataneni

Research Assistant/Associate (PhD): Fuel Cell Systems, PEM & SOFC

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SUMMARY

A highly motivated Automotive Engineering Master's student focused on **architecting fuel-cell system control** by merging system-level modeling with control theory for mobile hydrogen applications. An integrated contributor to DLR's aircraft fuel-cell system programme, **architecting and building a Model Predictive Control (MPC) approach** for the cathode-side control architecture of a 150-200 kW hydrogen-electric powertrain. Combines **leadership** of Germany's only student-built hydrogen fuel-cell vehicle team at Ecogenium with hands-on fuel-cell **test-bench validation** and industrial FCEV operating-strategy experience at Robert Bosch GmbH, backed by a growing scientific publication record including a peer-reviewed paper in review at SAE. Seeking a **Research Assistant/Associate PhD position** to apply this system-level modeling, control and test-bench expertise to fuel-cell systems research for mobile applications at RWTH's Chair of Thermodynamics of Mobile Energy Conversion Systems (TME).

PROFESSIONAL EXPERIENCE

Werkstudent, Fuel-Cell Systems (Aircraft) | DLR Institute for Small Aircraft Technologies (INK) 01/2026 – Present

Würselen (Aachen), Germany

Architected and built the complete cathode-side control architecture for the ~150-200 kW EnginePod fuel-cell system powering DLR's hydrogen-electric 9-seater commuter aircraft, combining a 0D lumped-parameter plant model with feed-forward, PI, decoupler and Model Predictive Control (MPC) strategies in MATLAB/Simulink, now calibrated against real system measurement data. Extended this modeling depth with a complementary 1D lumped model in GT Suite. Preceded this control work by engineering Python- and Excel-based sizing tools to select balance-of-plant components, deriving design specifications for the cathode, thermal-management and sensory systems and driving their procurement, then designing the complete system rig in CATIA V5 for packaging, piping, testing and component orientation.

Key Skills: MATLAB/Simulink, Model Predictive Control (MPC), GT Suite, 0D/1D Lumped-Parameter Modeling, Balance-of-Plant Sizing, CATIA V5, System Rig Design.

Mechanical / Suspension Team Lead & Pit Crew | Ecogenium e.V. (RWTH student team) 06/2024 – Present

Aachen, Germany

Leading fuel-cell system testing, integration and fault diagnosis this season on Germany's only student-built hydrogen fuel-cell vehicle, and architecting the energy-management and cooling-system control strategy for its newly built water-cooled stack in MATLAB/Simulink. Engineered, fabricated and electronically integrated a standalone fuel-cell test bench for the team's prior air-cooled stack, executing purging, humidity-cycling, short-circuiting and polarisation-curve experiments to map its optimal thermal operating window. Owns the fuel-cell system holistically, from component sizing and CAN-bus communication through control design to test and experimentation, a scope that helped secure a runner-up finish at Shell Eco-Marathon 2025.

Key Skills: Fuel-Cell Test-Bench Design, Polarisation Curves, Energy Management Strategy, MATLAB/Simulink, CAN Communication, Team Leadership.

Operating Strategy / System Simulation Intern | Robert Bosch GmbH 09/2025 – 12/2025

Schwieberdingen, Germany

Engineered a MATLAB-based powertrain strategy for Strong Fuel Cell Hybrid EVs that safeguards fuel-cell and HV-battery degradation, managing dynamic loads, thermal stress, low-voltage operation, SOC swing and life cycles through MAPs parameterized for different battery capacities and drive cycles. Developed and validated an iterative Simulink model of a rule-based power-split operating strategy in the (FC)EVsim PHEV module, correlating outputs against reference Excel and measurement data for high model fidelity, then executed multi-variable parametric optimization with Genetic Algorithms and Gradient Descent to generate control MAPs benchmarked against Dynamic Programming and ECMS. Formalized the resulting KPI analysis, covering power distribution, SOC window, efficiency and degradation, into a peer-reviewed manuscript now in review at SAE.

Key Skills: MATLAB/Simulink, Genetic Algorithms, Gradient Descent, Multi-Objective Optimization, DP/ECMS Benchmarking, Degradation Mitigation, Scientific Publication.

SKILLS

Fuel-cell systems & controls

PEM fuel-cell system architecture 0D lumped-parameter plant modeling 1D lumped modeling

feed-forward/PI/decoupler control design Model Predictive Control (MPC) back-pressure-valve (BPV) protection control

operating-strategy optimization balance-of-plant sizing & selection CAN communication

Modeling & simulation

MATLAB/Simulink GT Suite Dymola (Modelica) Genetic Algorithms Gradient Descent

DP/ECMS benchmarking

Test & measurement

fuel-cell test-bench build & commissioning single-cell & stack testing polarisation curves purging/humidity cycling

measurement-data evaluation & troubleshooting

Programming & data

Python (NumPy, Pandas) C/C++ Power BI

Design & CFD

CATIA V5 Siemens NX ANSYS (Fluent, Structural)

EDUCATION

M.Sc. Automotive Engineering, RWTH Aachen University

09/2023 – present

Aachen, Germany · GPA 1.7 (1.2 in core automotive courses) · Upcoming Master's thesis at DLR, direction: fuel-cell system control (scope in definition) · Relevant modules: Alternative & Mobile Propulsion Systems, Simulation Sciences, Mechatronics, Thermodynamics, FEM

B.Tech Mechanical Engineering, JNTU Hyderabad

06/2019 – 04/2023

Hyderabad, India · CGPA 9.05 (GPA 1.3, German scale)

PUBLICATIONS

- "A Comprehensive Analysis of Two Innovative Aircraft Design Configurations," IJSED Research, 11/2022.
- Co-author, peer-reviewed paper on a rule-based FCEV operating-strategy control approach (Robert Bosch GmbH), in review at SAE.

LANGUAGES

English (Fluent, C1) · German (A2-B1) · Telugu / Hindi (Native)